

Ubuntu Desktop Installation Using Proxmox Virtual Environment (PVE)

Introduction

Linux can be installed using different methods depending on the hardware and deployment requirements. Common installation methods include:

- **USB Drive Installation** – Uses a bootable USB flash drive to install Linux on a physical computer.
- **DVD/CD Installation** – Uses optical media to install Linux on a physical computer.
- **Network Installation (PXE Boot)** – Installs Linux over a network from a centralized server.
- **External Drive Installation** – Installs Linux on an external HDD or SSD.
- **Virtual Machine Installation** – Installs Linux inside virtualization platforms such as Proxmox VE, VMware, or VirtualBox.

For this laboratory exercise, **Ubuntu Desktop** will be installed using a **Virtual Machine (VM)** hosted on **Proxmox Virtual Environment (PVE)**. Virtualization provides a secure and flexible environment where students can learn Linux installation and administration without modifying physical computers.

Learning Objectives

At the end of this module, students should be able to:

1. Identify different Linux installation methods.
 2. Access the Proxmox VE web interface.
 3. Create an Ubuntu Desktop virtual machine.
 4. Install Ubuntu Desktop on a VM.
 5. Verify a successful installation.
 6. Perform basic Ubuntu Desktop operations.
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Materials Needed

Hardware

- Computer with a web browser
- Network connection

Software

- Proxmox Virtual Environment (PVE)
- Ubuntu Desktop ISO Image

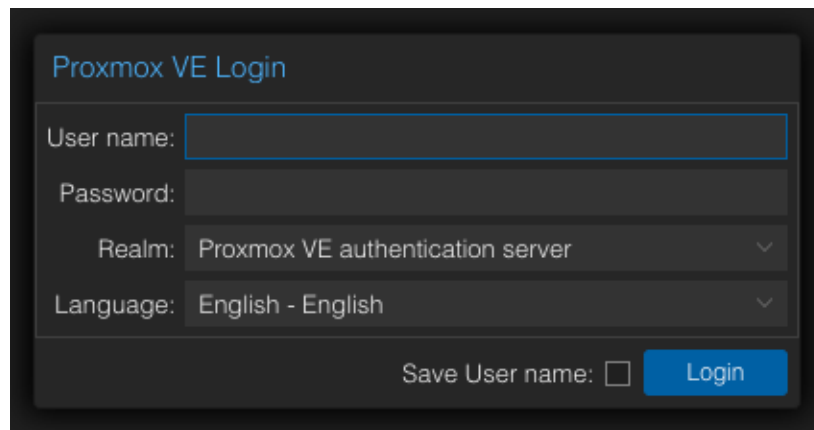
Accounts

- Proxmox VE user account

Laboratory Activity

Part I: Accessing Proxmox VE

Step 1: Login to Proxmox

A screenshot of the Proxmox VE Login web interface. The interface has a dark theme. At the top, it says "Proxmox VE Login" in a light blue font. Below this, there are four input fields: "User name:" with a text input box, "Password:" with a password input box, "Realm:" with a dropdown menu showing "Proxmox VE authentication server", and "Language:" with a dropdown menu showing "English - English". At the bottom right, there is a checkbox labeled "Save User name:" and a blue "Login" button.

1. Open a web browser.
2. Enter the Proxmox server URL: `https://<Proxmox-IP>:8006`
3. Enter your username and password.
4. Click **Login**

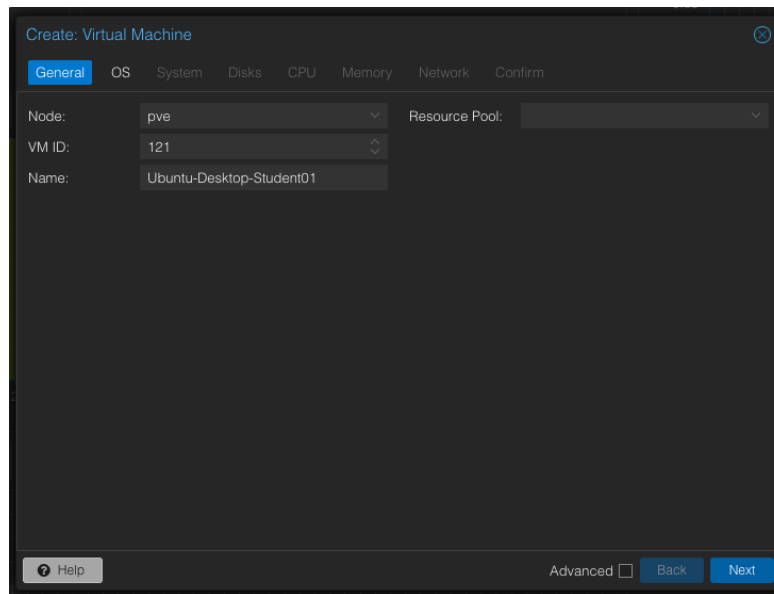
Expected Result

The Proxmox Dashboard is displayed.

Part II: Create an Ubuntu Desktop Virtual Machine

Step 1: Create VM

1. Click **Create VM**.
2. Set VM ID: 121
3. Set: Name: Ubuntu-Desktop-Student01
4. Click Next.



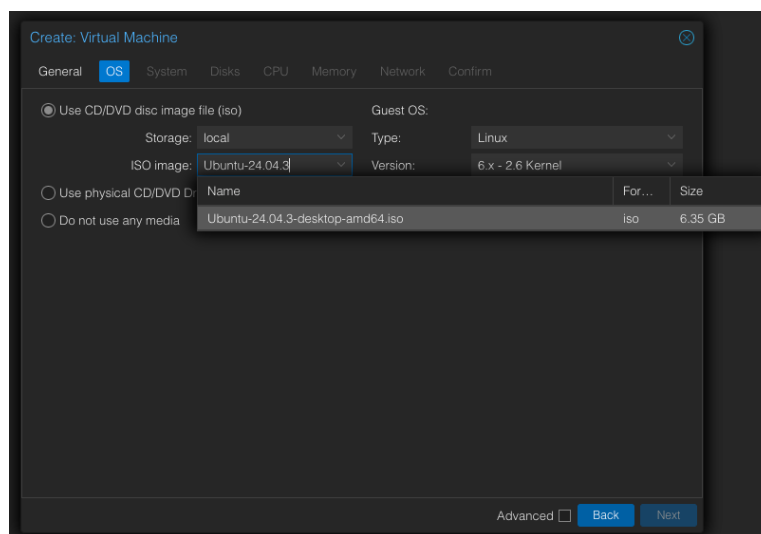
The screenshot shows the 'Create: Virtual Machine' dialog box with the 'General' tab selected. The fields are filled as follows:

Field	Value
Node	pve
VM ID	121
Name	Ubuntu-Desktop-Student01
Resource Pool	

At the bottom, there is a 'Help' button, an 'Advanced' checkbox, and 'Back' and 'Next' buttons.

Step 2: Select Ubuntu ISO

1. Choose the provided Ubuntu Desktop ISO.
2. Click **Next**.



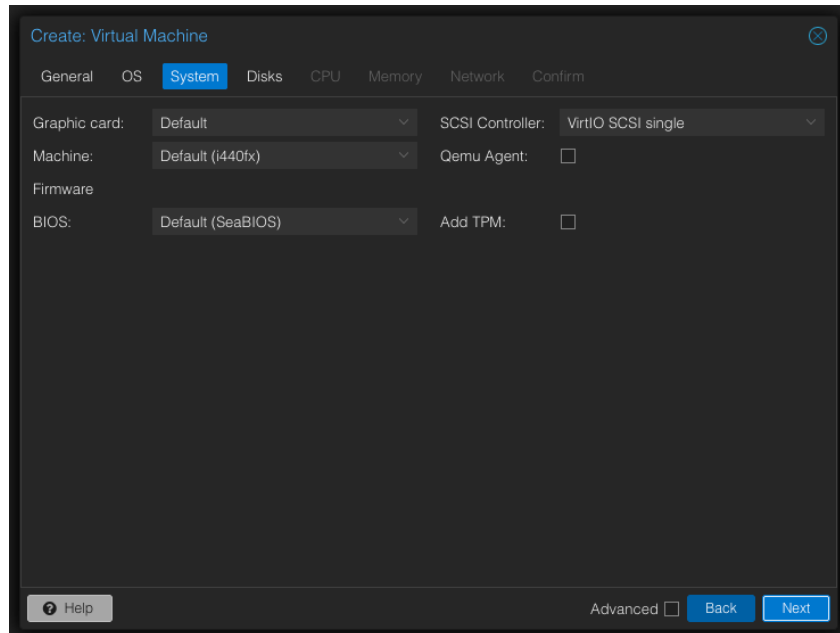
The screenshot shows the 'Create: Virtual Machine' dialog box with the 'OS' tab selected. The 'Use CD/DVD disc image file (iso)' option is selected. The 'Guest OS' section shows 'Type: Linux' and 'Version: 6.x - 2.6 Kernel'. The 'ISO image' dropdown is open, showing a list of ISO files.

Name	For...	Size
Ubuntu-24.04.3-desktop-amd64.iso	iso	6.35 GB

At the bottom, there is an 'Advanced' checkbox and 'Back' and 'Next' buttons.

Step 3: Configure System

1. Use the default settings.
2. Click **Next**.



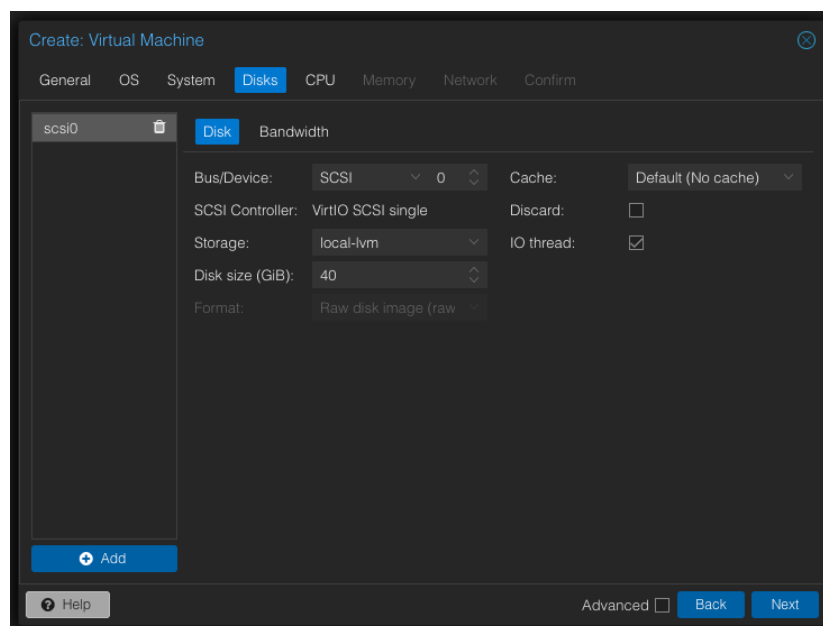
The screenshot shows the 'Create: Virtual Machine' window with the 'System' tab selected. The window has a dark theme and a close button in the top right corner. The tabs at the top are General, OS, System (selected), Disks, CPU, Memory, Network, and Confirm. The System tab contains the following settings:

Setting	Value
Graphic card:	Default
Machine:	Default (i440fx)
Firmware	
BIOS:	Default (SeaBIOS)
SCSI Controller:	VirtIO SCSI single
Qemu Agent:	☐
Add TPM:	☐

At the bottom, there is a 'Help' button, an 'Advanced' checkbox, and 'Back' and 'Next' buttons.

Step 4: Configure Storage

1. Set: Disk Size: 40 GB
2. Click **Next**.



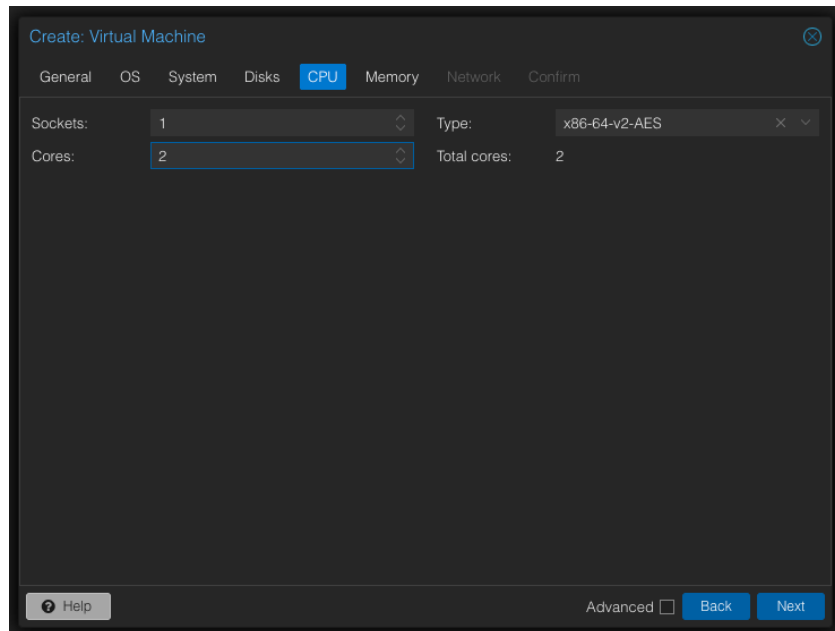
The screenshot shows the 'Create: Virtual Machine' window with the 'Disks' tab selected. The window has a dark theme and a close button in the top right corner. The tabs at the top are General, OS, System, Disks (selected), CPU, Memory, Network, and Confirm. The Disks tab contains the following settings:

Setting	Value
Bus/Device:	SCSI 0
Cache:	Default (No cache)
SCSI Controller:	VirtIO SCSI single
Storage:	local-lvm
Disk size (GiB):	40
Format:	Raw disk image (raw)
Discard:	☐
IO thread:	☒

At the bottom, there is an 'Add' button, a 'Help' button, an 'Advanced' checkbox, and 'Back' and 'Next' buttons.

Step 5: Configure CPU

1. Set: CPU Cores: 2
2. Click **Next**.

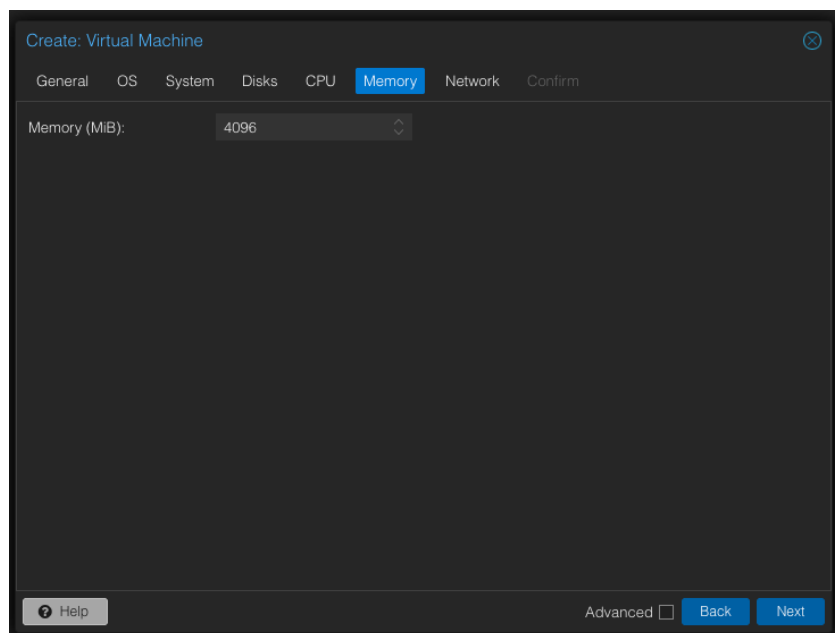


The screenshot shows the 'Create: Virtual Machine' dialog box with the 'CPU' tab selected. The 'Sockets' dropdown is set to '1' and the 'Cores' dropdown is set to '2'. The 'Type' dropdown is set to 'x86-64-v2-AES'. The 'Total cores' is displayed as '2'. At the bottom, there is a 'Help' button, an 'Advanced' checkbox, and 'Back' and 'Next' buttons.

Property	Value
Sockets	1
Cores	2
Type	x86-64-v2-AES
Total cores	2

Step 6: Configure Memory

1. Set: Memory: 4096 MB (4 GB)
2. Click **Next**.



The screenshot shows the 'Create: Virtual Machine' dialog box with the 'Memory' tab selected. The 'Memory (MiB)' dropdown is set to '4096'. At the bottom, there is a 'Help' button, an 'Advanced' checkbox, and 'Back' and 'Next' buttons.

Property	Value
Memory (MiB)	4096

Step 7: Configure Network

1. Set: Bridge: vmbr1
2. Uncheck Firewall
3. Click **Next**.

The screenshot shows the 'Create: Virtual Machine' dialog box with the 'Network' tab selected. The 'General' tab is also visible. The 'No network device' checkbox is unchecked. The 'Bridge' is set to 'vmbr1', 'Model' is 'VirtIO (paravirtualized)', 'VLAN Tag' is 'no VLAN', and 'MAC address' is 'auto'. The 'Firewall' checkbox is unchecked. At the bottom, there is a 'Help' button, an 'Advanced' checkbox, and 'Back' and 'Next' buttons.

Key	Value
cores	2
cpu	x86-64-v2-AES
ide2	local:iso/Ubuntu-24.04.3-desktop-amd64.iso.media=cdrom
memory	4096
name	Ubuntu-Desktop-Student01
net0	virtio,bridge=vmbr1
nodename	pve
numa	0
ostype	l26
scsi0	local-lvm:40,iothread=on
scsihw	virtio-scsi-single
sockets	1
vmid	121

Step 8: Complete VM Creation

1. Review the configuration.
2. Click **Finish**.

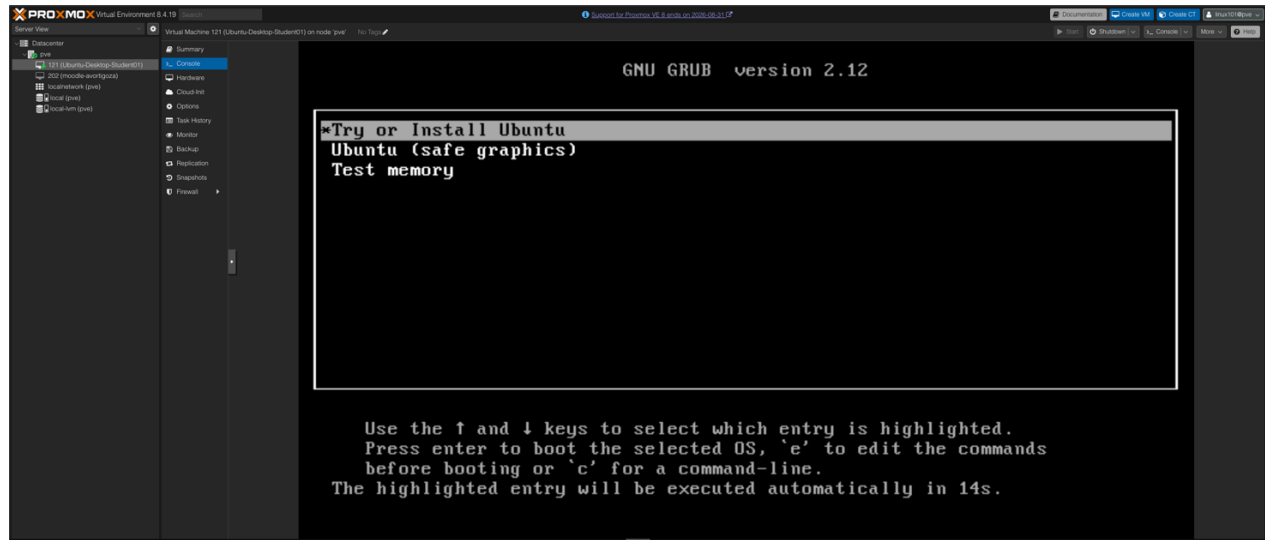
The screenshot shows the 'Create: Virtual Machine' dialog box with the 'Confirm' tab selected. The 'General' tab is also visible. The 'Confirm' tab displays a table of configuration keys and values. At the bottom, there is a 'Start after created' checkbox, an 'Advanced' checkbox, and 'Back' and 'Finish' buttons.

Key	Value
cores	2
cpu	x86-64-v2-AES
ide2	local:iso/Ubuntu-24.04.3-desktop-amd64.iso.media=cdrom
memory	4096
name	Ubuntu-Desktop-Student01
net0	virtio,bridge=vmbr1
nodename	pve
numa	0
ostype	l26
scsi0	local-lvm:40,iothread=on
scsihw	virtio-scsi-single
sockets	1
vmid	121

Part III: Install Ubuntu Desktop

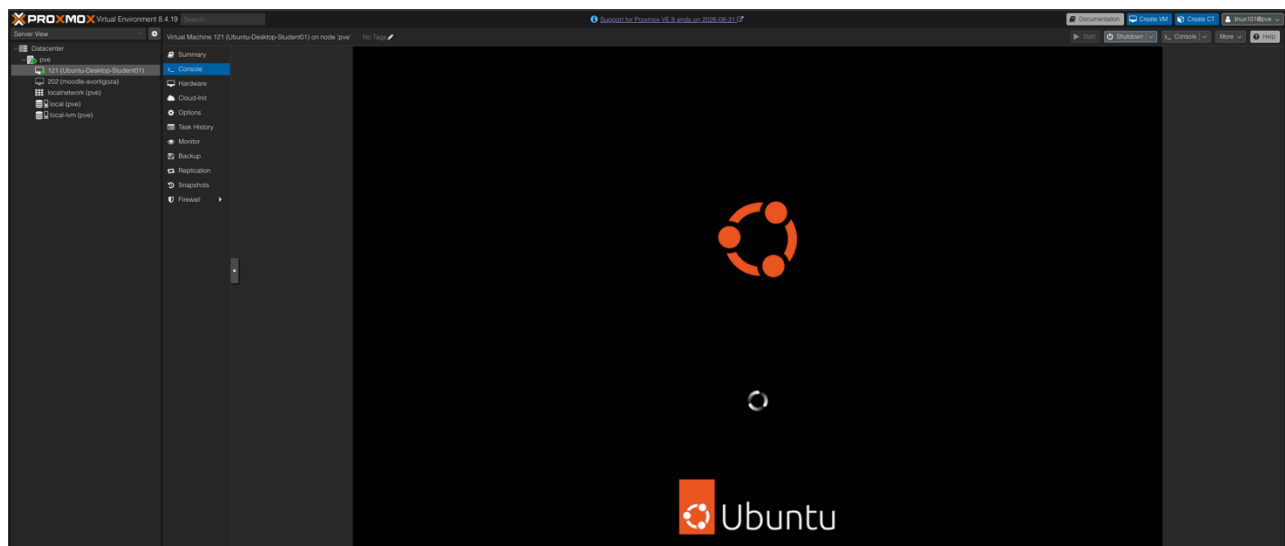
Step 1: Start the VM

1. Select the VM.
2. Click **Start**.
3. Open the **Console** tab.



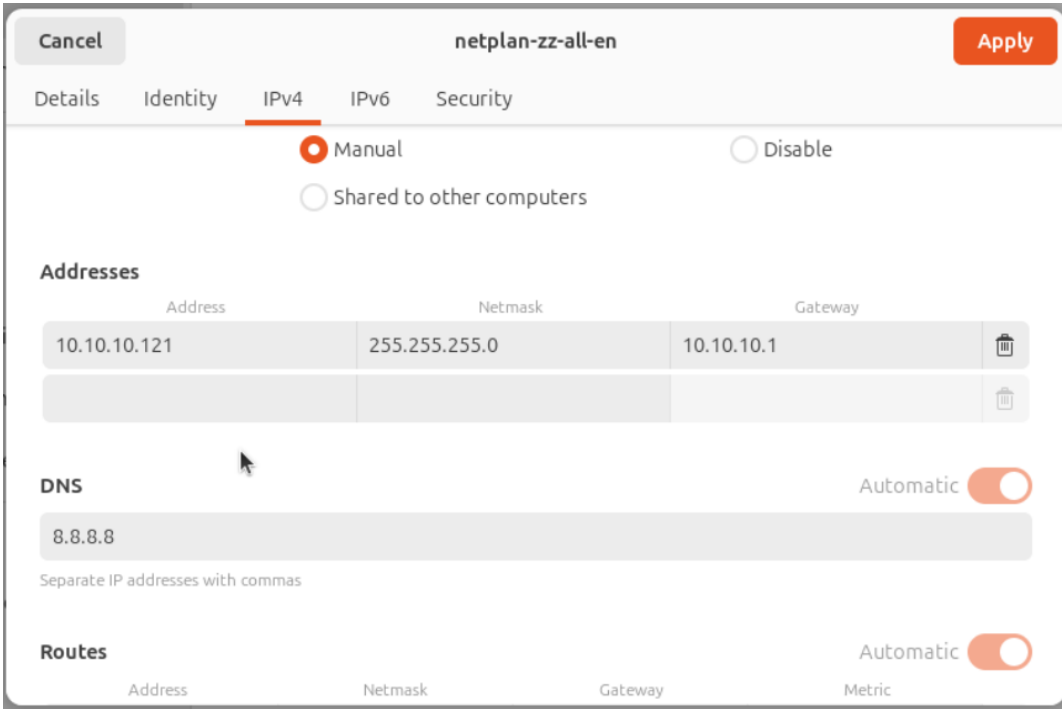
Step 2: Launch Ubuntu Installer

1. Select **Try or Install Ubuntu**.
2. Wait for the Ubuntu Desktop environment to load.



Step 3: Choose Preferences

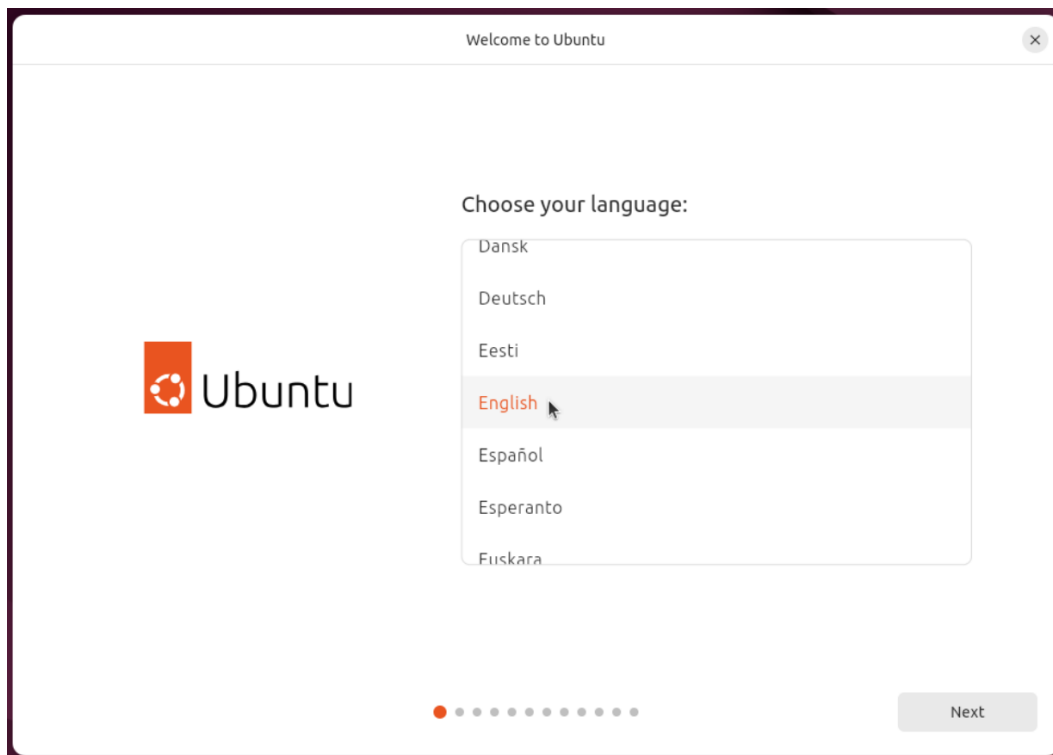
1. Select your language.
2. Choose your keyboard layout.
3. Connect to the network if available.
4. Click **Next**.



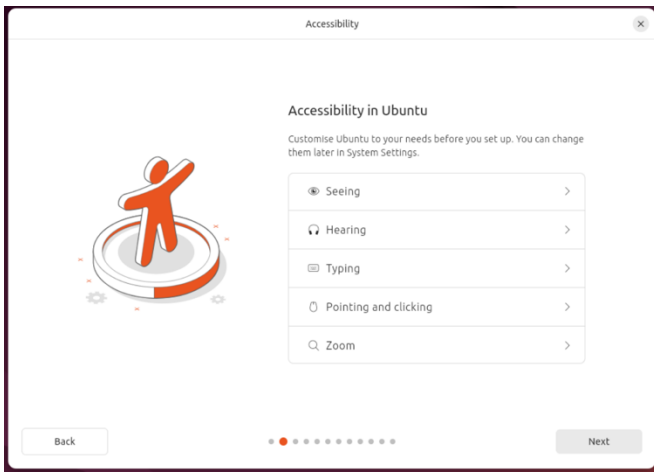
The image shows the 'netplan-zz-all-en' network configuration window. It has tabs for 'Details', 'Identity', 'IPv4', 'IPv6', and 'Security'. The 'IPv4' tab is selected. Under 'IPv4', there are two radio buttons: 'Manual' (selected) and 'Disable'. Below that is a radio button for 'Shared to other computers'. The 'Addresses' section has a table with columns 'Address', 'Netmask', and 'Gateway'. The first row contains '10.10.10.121', '255.255.255.0', and '10.10.10.1'. The 'DNS' section has a text input field containing '8.8.8.8' and a toggle switch for 'Automatic' which is turned on. The 'Routes' section has a table with columns 'Address', 'Netmask', 'Gateway', and 'Metric', and a toggle switch for 'Automatic' which is also turned on.

Address	Netmask	Gateway
10.10.10.121	255.255.255.0	10.10.10.1

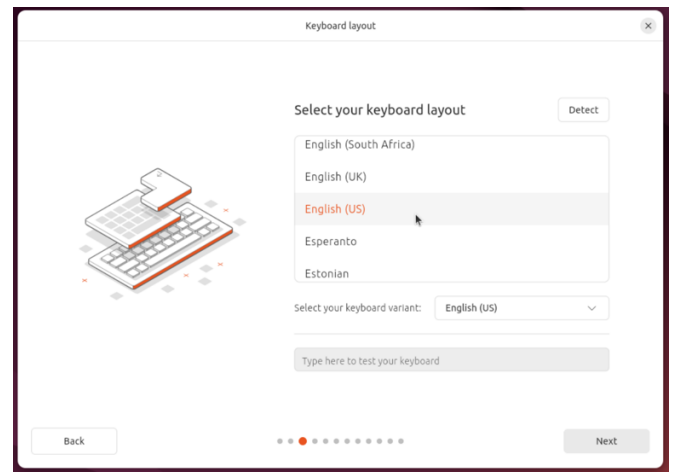
Address	Netmask	Gateway	Metric



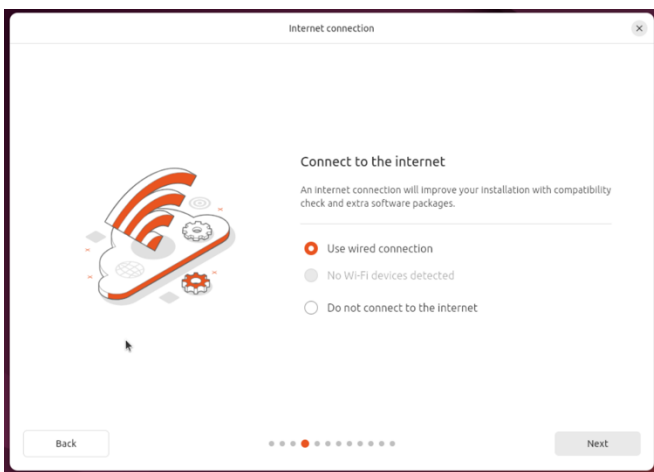
The image shows the 'Welcome to Ubuntu' window. It has the Ubuntu logo on the left. On the right, it says 'Choose your language:' followed by a list of languages: Dansk, Deutsch, Eesti, English (highlighted with a mouse cursor), Español, Esperanto, and Fuskara. At the bottom right is a 'Next' button. At the bottom center is a progress indicator with 10 dots, the first of which is filled.



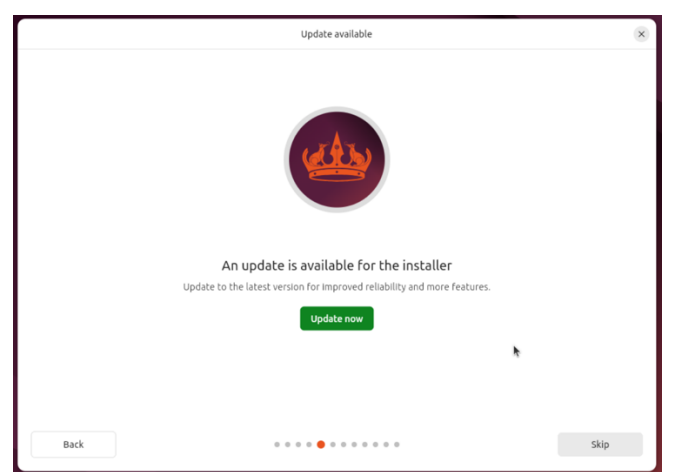
1



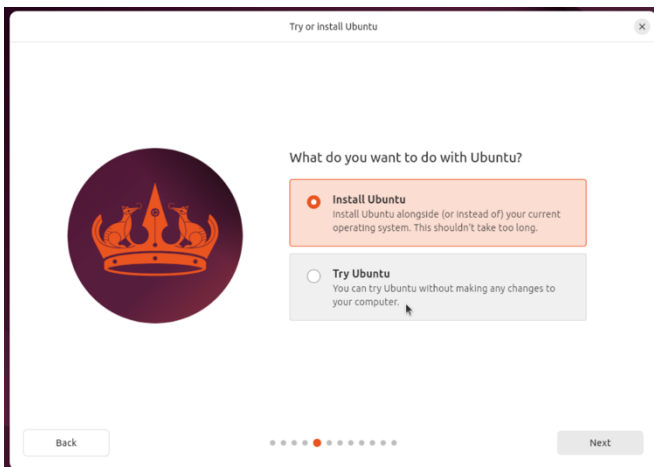
2



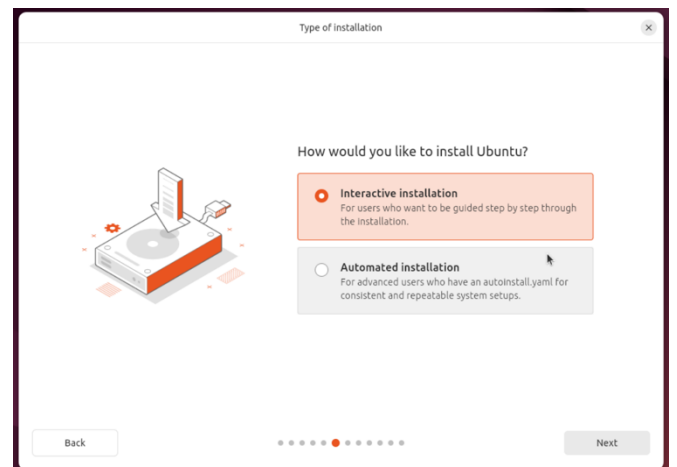
3



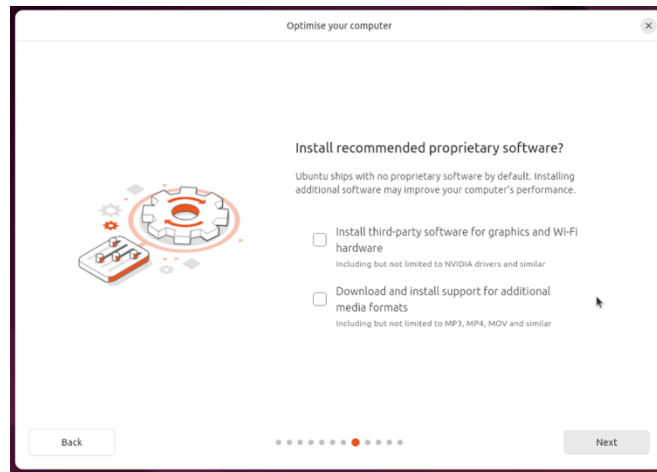
4



5



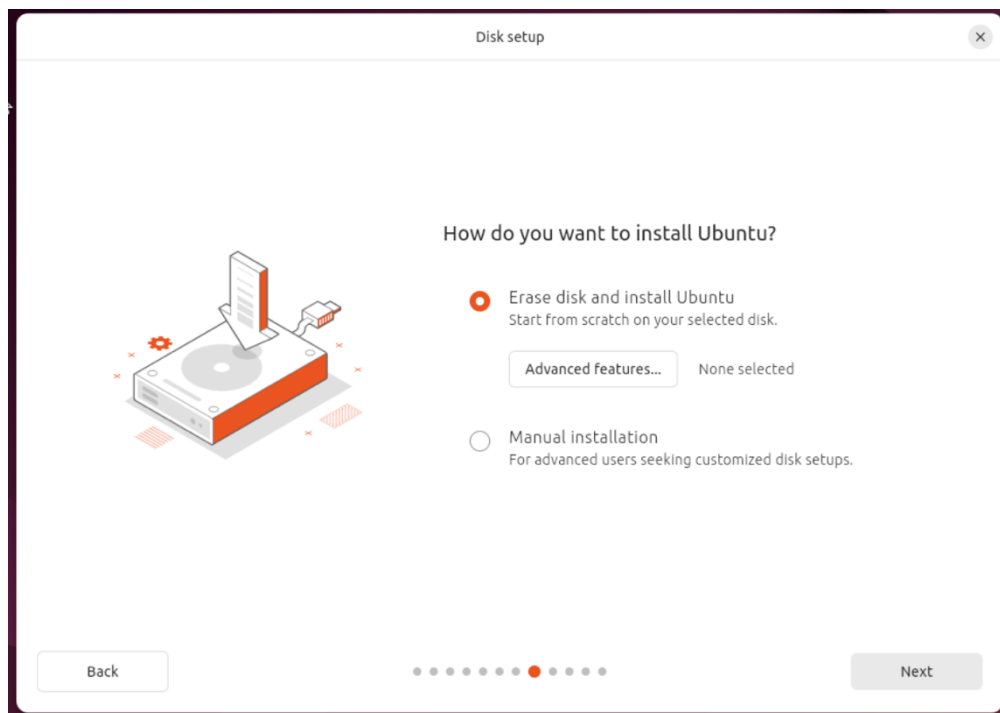
6



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Step 4: Installation Type

1. Select: Erase disk and install Ubuntu
2. Click Next.



Step 5: Create User Account

1. Your Name: Student Name
2. Computer Name: Ubuntu-Desktop-Lab01
3. Username: student01
4. Password: password123
5. Click Next.

Create your account

Create your account

Your name
Eugene Horfilla ✓

Your computer's name
Ubuntu-Desktop-Lab01 ✓

Your username
student01 ✓

Password
password123 Hide Weak password

Confirm password
password123 ✓

☒ Require my password to log in

☐ Use Active Directory

Back Next

Step 6: Select Time zone

1. Set Location and Time zone
2. Click Next.

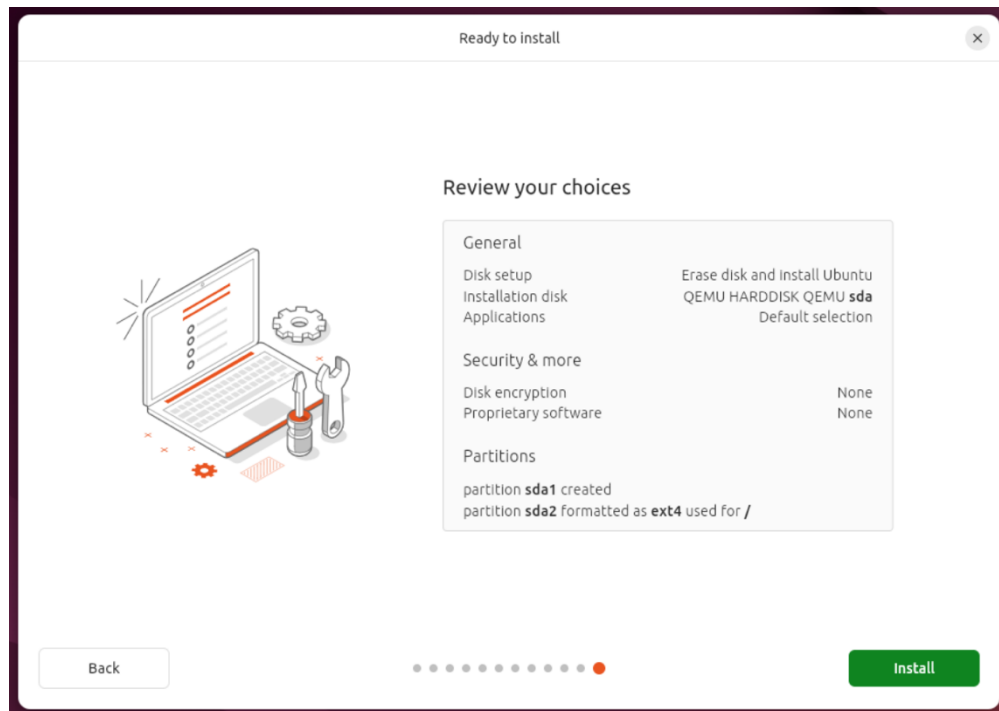
Select your timezone

Location
Quezon City (Metro Manila, Philippines)

Timezone
Asia/Manila

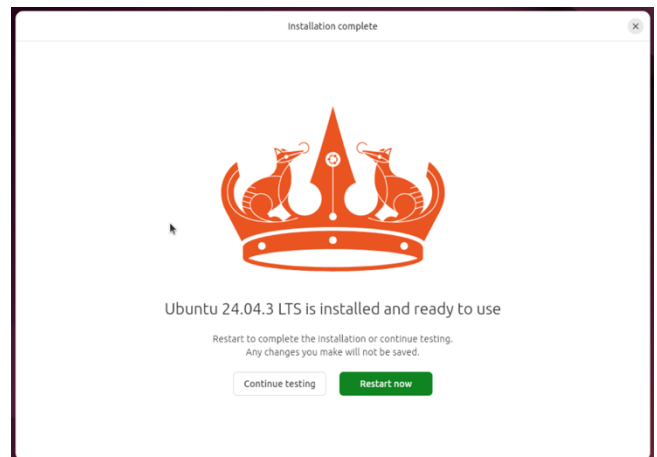
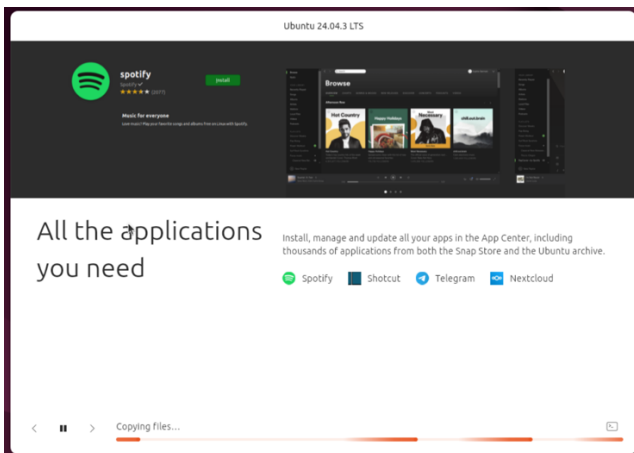
Back Next

Step 7: Review and Install



Step 8: Finish Installation

1. Wait for Ubuntu Desktop to install.
2. Click **Restart Now** when prompted.
3. Allow the VM to reboot.



Part IV: Post-Installation Verification

Verify Desktop Access

1. Log in using the account created during installation.
2. Confirm that the Ubuntu Desktop interface loads successfully.

Verify Ubuntu Version

Open **Terminal** and run:

Verify OS:	<code>cat /etc/os-release</code>
Verify Hostname:	<code>hostname</code>
Verify IP Address:	<code>ip addr</code>
Verify Disk Space:	<code>df -h</code>
Update Ubuntu:	<code>sudo apt update</code> <code>sudo apt upgrade -y</code>

Part V: Basic Ubuntu Desktop Familiarization

Activity 1: Open Applications

1. Click **Show Applications**.
2. Open:
 - Firefox
 - Terminal
 - Files

Activity 2: Create a Folder

1. Open **Files**.
2. Navigate to **Home**.
3. Create a folder named: LinuxLab

Activity 3: Create a Text File

1. Open **Text Editor**.
2. Type: Ubuntu Desktop Installation Completed Successfully.
3. Save the file inside the **LinuxLab** folder.

